

Title:

EV Energy Consumption Prediction Regression Techniques

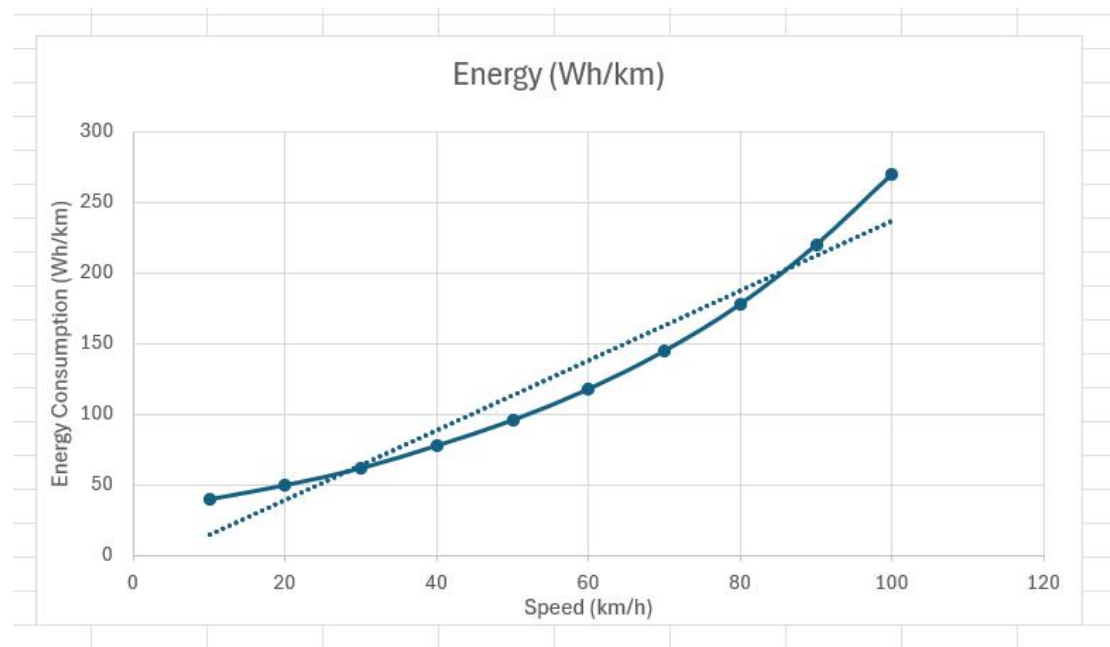
Objective:
To predict EV energy consumption based on vehicle speed.

Dataset:
Synthetic EV speed and energy consumption data.

1.Linear Regression

Methodology:

1. Created dataset in Excel.
2. Plotted Speed vs Energy Consumption graph.
3. Applied Linear Regression trendline.
4. Analyzed the relationship between speed and energy consumption.



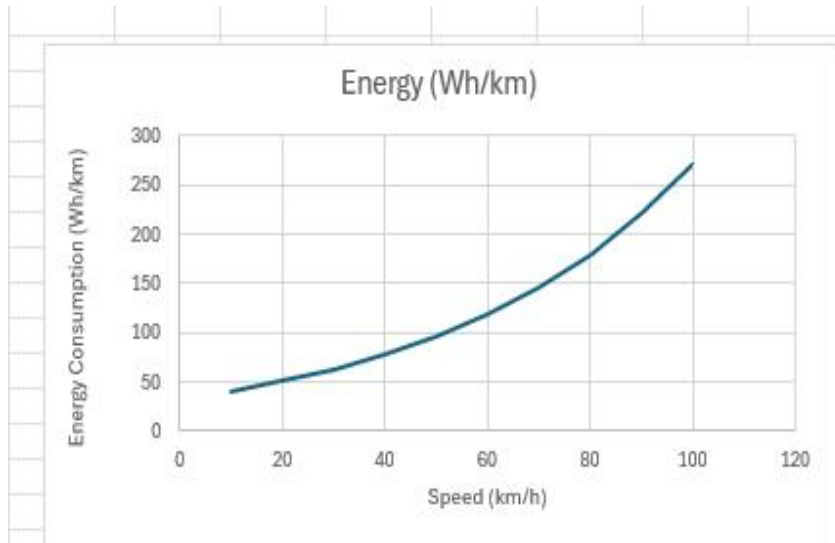
Observation:

1. Energy consumption increases as vehicle speed increases.
2. The trendline is a straight line.
3. Linear Regression is simple and easy to implement.
4. Suitable for identifying overall trends in EV data.

2 .Polynomial Regression

Methodology

1. Collected vehicle speed and energy consumption data.
2. Plotted Speed vs Energy Consumption graph in Excel.
3. Applied a Linear Regression trendline.
4. Analyzed the relationship between speed and energy consumption.



Observation:

- Energy consumption does not increase perfectly linearly at higher speeds.
- Polynomial Regression produces a curved trendline.
- The curve better represents nonlinear EV behavior.
- More suitable for accurate prediction at higher vehicle speeds.

Conclusion;

1. Both regression techniques can predict EV energy consumption.
2. Linear Regression is simple and easy to understand.
3. Polynomial Regression captures nonlinear trends more effectively.
4. Polynomial Regression generally provides better prediction accuracy for EV energy consumption data.

Tools Used:
Microsoft Excel

Data_Table

Speed	Energy (Wh/km)
10	40
20	50
30	62
40	78
50	96
60	118
70	145
80	178
90	220
100	270